



Low-Cost Acoustic Localization for Small-Scale Underwater Vehicles: Motion-Constrained Path-Planning

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Motivation

- Localization crucial for robot operation
- No GNSS available underwater
- Vision based localization restricted to clear water
- Acoustic localization robust against turbidity

But: Low-cost acoustic modems have limited range and slow measurement rates [1]

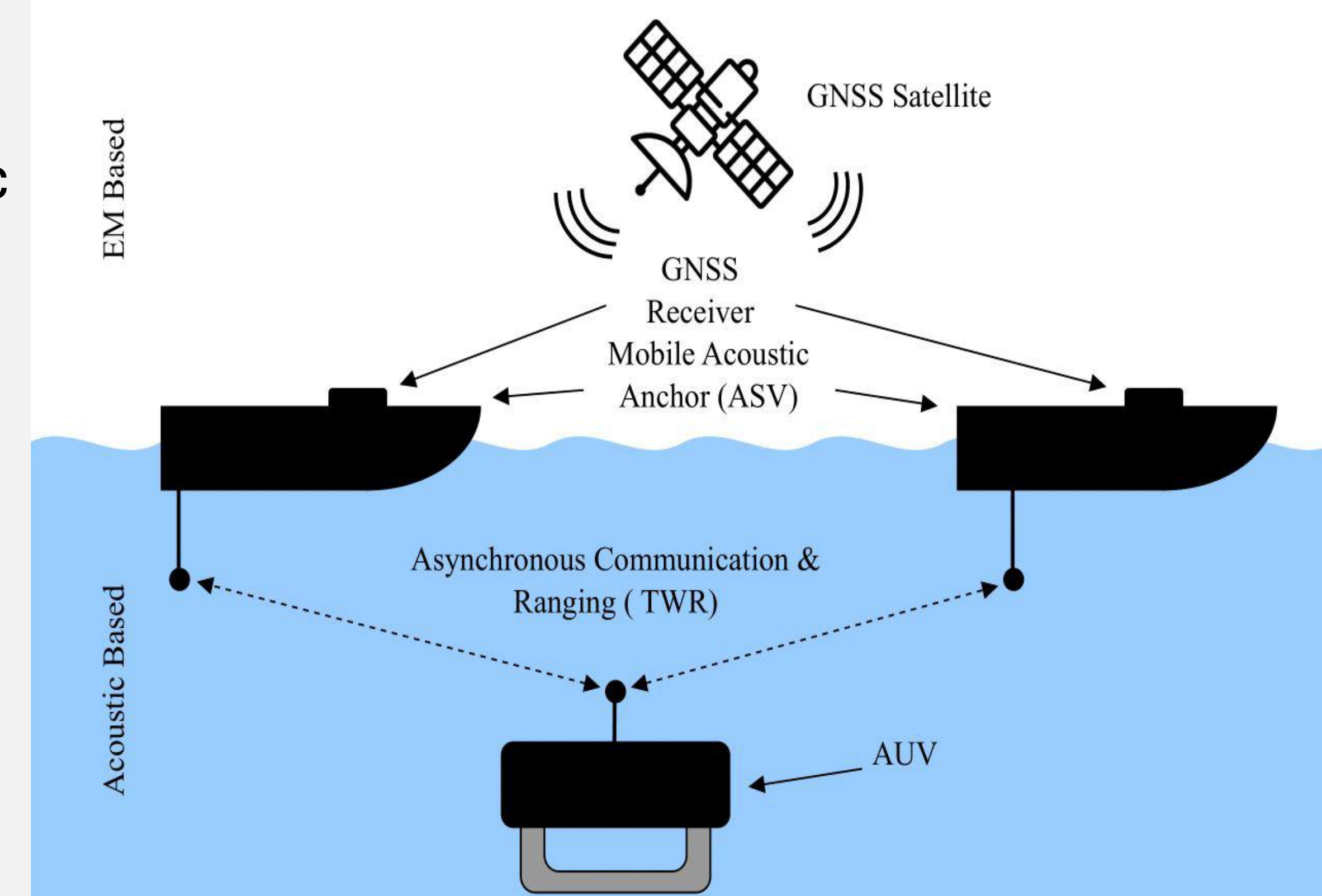
→ **Mobile Acoustic Anchors**

Goal

How to position Mobile Acoustic Anchors to achieve the best localization result for an underwater vehicle?

Key Challenges

- Incomplete and noisy measurements
- Nonlinear system dynamics

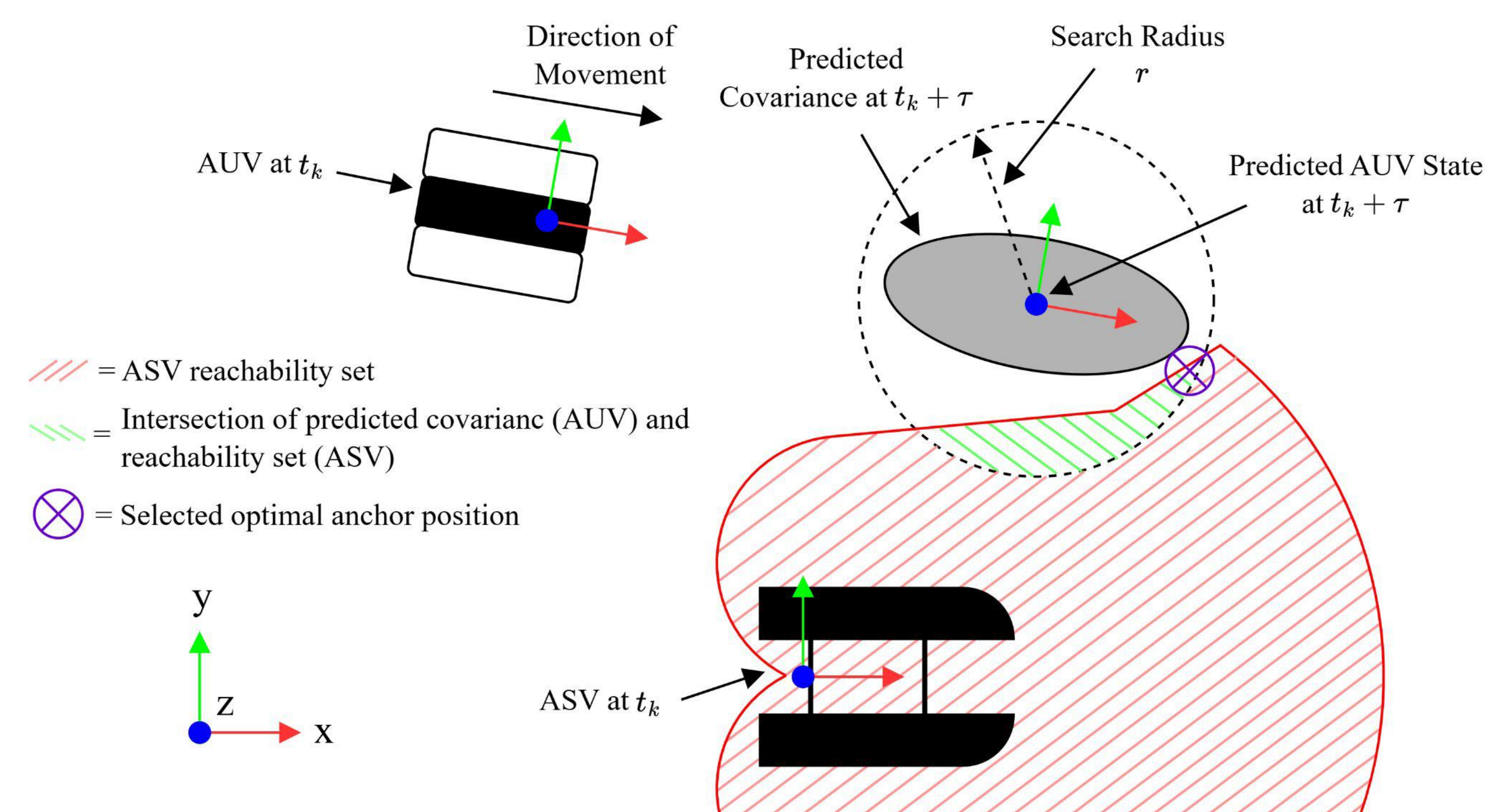


Concept

- Predict AUV position with covariance
- Calculate optimal next ASV position for measurement
- Minimize covariance of AUV position estimate
- Restrict optimization to ensure reachability of ASV

→ **Motion-constrained Cordinated Path Planning**

Advantage: Predicted information gain in filter is actually realized because the ASV reaches the measurement position



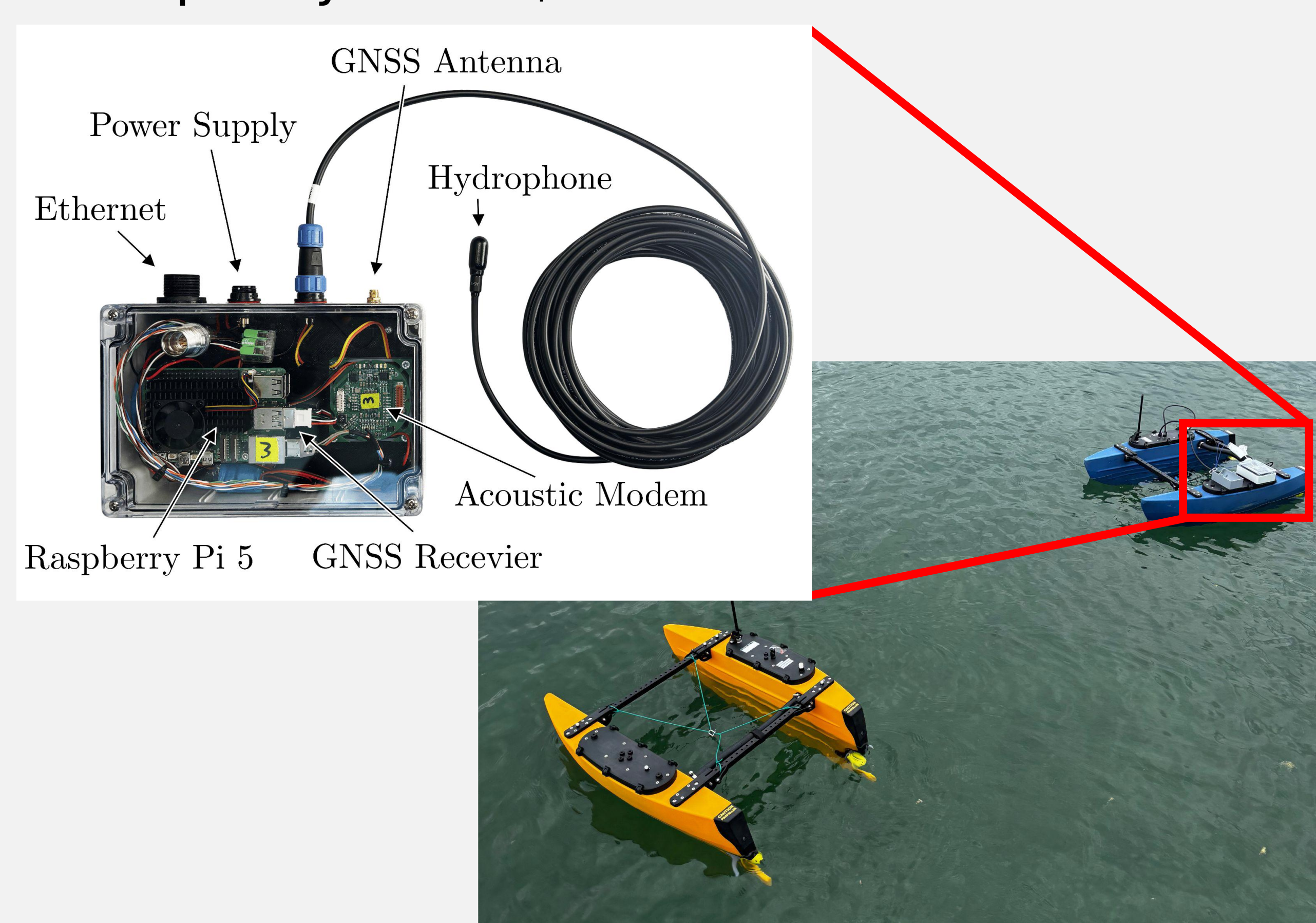
Field Experiments

- Experiments on small lake (Hollerner See: 48.284791° N, 11.606319° E)
- Target trajectory 20 x 20 m square
- Blue ASV: Mobile Acoustic Anchor
- Orange ASV: Mimics underwater vehicle, speed of 0.4 m/s

→ **Ground truth from RTK GNSS**

Hardware

- Ahoi open source acoustic modem 200\$ [2]
- Aquarian AS-1 acoustic hydrophone 400\$
- Raspberry Pi 5 80\$

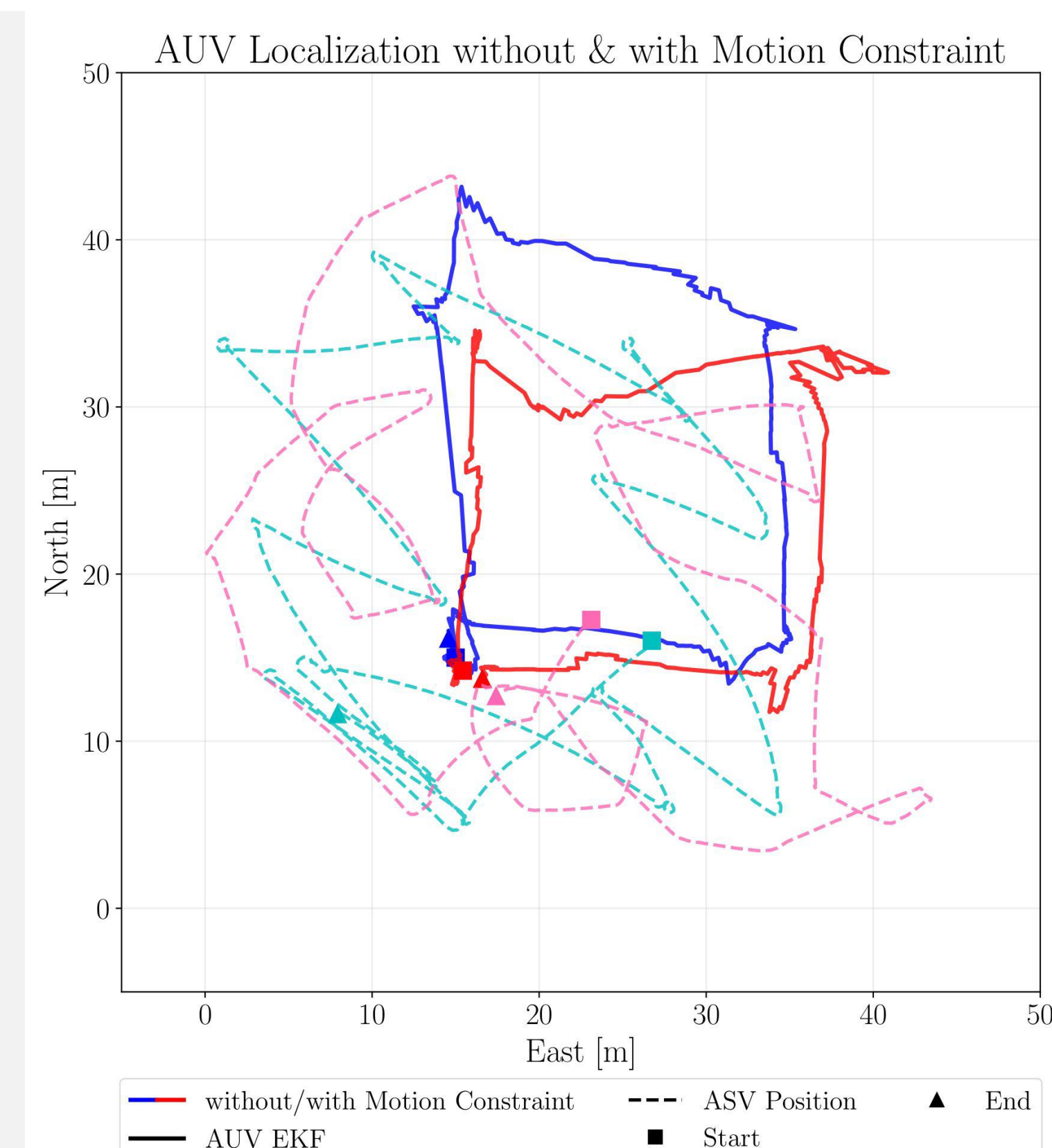


Setup for experimental evaluation using two BlueBoats. The hardware is integrated into a weather-resistant sensor box.

Results

- Evaluation on 10 runs:
 - *Baseline*: 5x Coordinated Path Planning (CPP) [3]
 - *Ours*: 5x Motion-constrained CPP
- Results vary between runs

	Median	RMS
Ours	0.73 m	1.79 m
Baseline	0.8 m	1.88 m



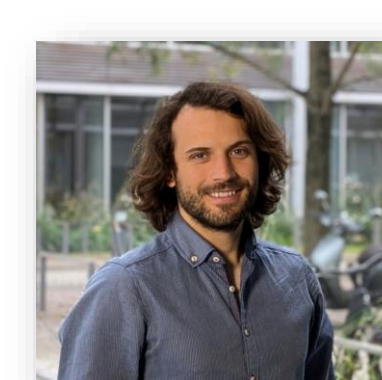
Comparison of localization results between motion-constrained and unconstrained CPP

Conclusion & Outlook

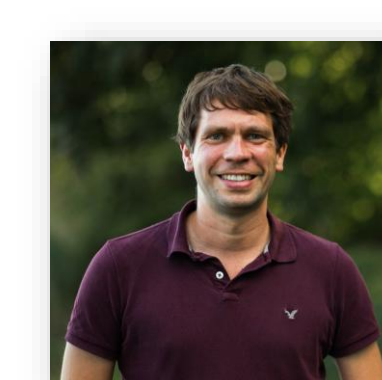
- Approach shows on-par performance in practice
- To increase robustness, accuracy and allow higher speeds a second ASV is required
- Implement factor graph for smoothing to improve overall accuracy [4]

- [1] D. A. Duecker, F. Steinmetz, E. Kreuzer and C. Renner, "Micro AUV Localization for Agile Navigation with Low-cost Acoustic Modems," 2020 IEEE/OES Autonomous Underwater Vehicles Symposium (AUV), St. Johns, NL, Canada, 2020
- [2] Bernd-Christian Renner, Jan Heitmann, and Fabian Steinmetz, "Ahoi: Inexpensive, Low-power Communication and Localization for Underwater Sensor Networks and μ AUVs," ACM Trans. Sen. Netw. 16, 2020
- [3] A. Quraishi and A. Martinoli, "Coordinated Path Planning for Surface Acoustic Beacons for Supporting Underwater Localization," 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Prague, Czech Republic, 2021
- [4] R. Turrisi, D. A. Duecker, J. Morrison, F. Steinmetz and M. Benjamin, "ASV-Aided AUV Navigation: A Field Study on Nonlinear Estimation for Localization of Low-Cost, Scalable Systems," 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, 2025

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